

LAST (Family) NAME: _____

FIRST (Given) NAME: _____

STUDENT NUMBER: _____

UNIVERSITY OF TORONTO
Faculty of Arts & Science

June 2022 EXAMINATIONS

ECO372H1F

Data Analysis and Applied Econometrics in Practice

Duration: 2 hours
Aids Allowed: None

Exam Reminders:

- Fill out your name and student number on the top of this page.
- If a scantron and/or exam booklets are required: Ensure you fill in your name and student number on the scantron and/or exam booklet(s)
- Do not begin writing the actual exam until the announcements have ended and the Exam Facilitator has started the exam.
- As a student, you help create a fair and inclusive writing environment. If you possess an unauthorized aid during an exam, you may be charged with an academic offence.
- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- When you are done your exam, raise your hand for someone to come and collect your exam. Do not collect your bag and jacket before your exam is handed in.
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.
- In the event of a fire alarm, do not check your cell phone when escorted outside.

Special Instructions:

- All answers should be written on the exam paper itself.

Exam Format and Grading Scheme:

Section I: 35 marks
Section II: 30 marks
Section III: 20 marks

Students must hand in all examination materials at the end

Section 1: Evaluation of a Campaign Against Gender-Based and Intimate Partner Violence (35 points)

The Secretariat of Women of Mexico City (SEMUJERES CDMX) has a policy priority during the term 2019-2022 to decrease the incidence of intimate partner violence (IPV) and gender-based violence (GBV) across the jurisdiction. A first programmatic effort of SEMUJERES involved a door-to-door campaign carried out in 2020 across neighborhoods (“colonias”) considered to be of high incidence of these forms of violence based on data from years 2017-18. In this program:

- (a) Female community health volunteers (CHVs) visited the census of households during the early evening and ask to speak in private with an adult woman in the home;
- (b) The CHV would deliver a booklet highlighting (i) women’s rights to a life without intimate partner or other forms of gender-based violence and (ii) contact information of public or civil sector organizations that could assist the person in case she needed support services as a result of (the risk of) victimization.

Policy Question / Evaluation: The Secretariat requested that its research and evaluation unit (R&E Unit) conduct a quantitative analysis to estimate the impact of such efforts *one (1) year* following the door-to-door campaign on available measures of services usage and victim protection.

Data: Specifically, the Secretariat has access to *administrative microdata* on individuals’:

- (1) visits to public and NGO specialized support services accessible in her neighborhood;
- (2) restraining orders issued by state CDMX tribunals against offenders found to be engaging in GBV against an individual who had reported this to law enforcement and judicial authorities;
- (3) the age, gender, neighborhood (“colonia”) of residence, and geo-referenced residential address of the individual (victim or potential victim).

Note: Assume for simplicity that we have a discrete treatment, $D_{ihnt} = \{0,1\}$, whether the door-to-door campaign reaches individual i of household h in neighborhood n at time period t ; vector of observed outcomes $\mathbf{Y}_{ihnt}$, and exogenous covariates $\mathbf{X}_{ihn}$.

Each one of the two (2) potential outcomes from the $\mathbf{Y}_{ihnt}$, are defined as Y_{1i} , Y_{0i} , and the observed outcome is defined following the standard switching equation.

Part A. The Secretary wishes that the R&E Unit estimate the *causal effect* of the campaign, using a linear probability model framework for each one of the two outcomes above along the following lines:

$$Y_{ihnt} = \alpha + \theta D_{ihnt} + \mathbf{X}_{ihn}\beta + u_i$$

- With specific reference to the formula for omitted variables and selection bias, explain the potential problem in using OLS to estimate the causal ATT parameter, θ . [10]

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Part B. Given the access to geo-referenced residential location data for individuals, the R&E Unit considers comparing outcomes of individuals who along the borders of the targeted neighborhoods (“colonias”) to those who live in nearby neighborhoods outside of the targeted areas.

- Explain why this strategy is (in principle) a Regression Discontinuity Design [5]:
(NB: No need to write down equations here; see next question)

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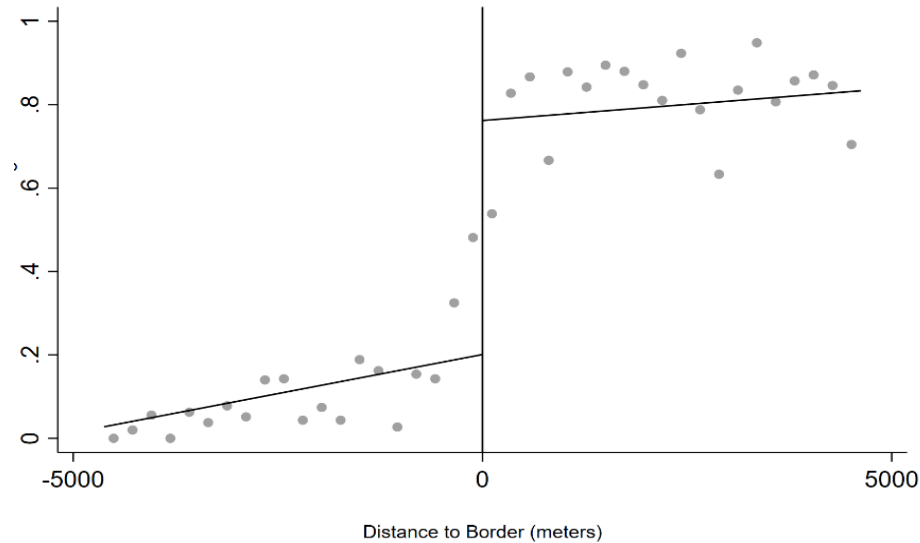
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Part B (continued):

- Discuss two valid implementations of this Regression Discontinuity Design-based strategy (write down the empirical models), and the conditions under which it will yield unbiased (consistent) estimates of “the” treatment effect θ . You should discuss the ability of this strategy to address the selection problem [10]:

Part C: The SEMUJERES R&E Unit generates RDD estimates of the causal effects of the campaign on the number of residents who access specialized support services (see Figure 1).

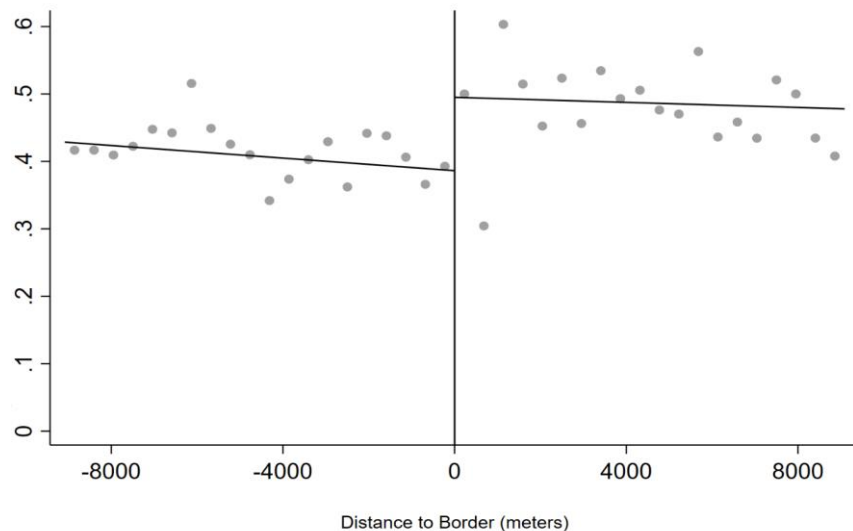
Figure 1: Number of Residents who Access Specialized Support Services



Coefficient: 0.497, p-value = 0.000

Notes: The dots represent local averages of the total number of residents who visit specialized support services by distance to “colonia” borders. Local linear fit using Calonico et al. (2016) optimal bandwidth and local linear regression RD estimation.

Figure 2: Number of Restraining Orders Issued by Judicial Authorities against Offenders



Coefficient: 0.090, p-value = 0.031

Notes: The dots represent local averages of the total number of restraining orders issued by CDMX state tribunals by distance of the victim’s residence to the “colonia” borders. Local linear fit using Calonico et al. (2016) optimal bandwidth and local linear regression RD estimation.

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- With reference to your answers to the previous questions in this section, critically discuss their results. In your role as analyst, emphasize the interpretation of the results, threats to identification, and necessary tests to verify the validity of the design for purposes of this evaluation [10]:

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Section 2: Instrumental Variables (30 points)

Part A: Following the presentation of the results from Section 1 (above) to SEMUJERES CDMX, the Secretariat requests an analysis to understand how individuals' *visits* to public and NGO specialized support services accessible in the neighborhood can affect the probability that victims are able to gain protection through restraining orders issued by state CDMX tribunals against offenders.

The R&E Unit team considers implementing an instrumental variables (IV) research design to aim to address this question, using the targeted neighborhood visits (Z_{nt}) as an IV for the take-up of specialized services (D_{ihnt}) and relying on the variation along the borders of treated and comparison neighborhoods.

- Explain why this strategy is (in principle) a Fuzzy Regression Discontinuity Design. Present to the research team the empirical models to be estimated to implement this IV strategy that relies on the geographic discontinuity **[10]**:
[NB: There is no need to discuss validity of interpretation of results, see next question]

Part B: You estimate the IV/Fuzzy RDD Model and generate the following output:

Table 1: IV Estimate of Effect of Specialized Services Usage on Issue of Restraint Orders
(among Requesters)

Dependent variable: Issue of Restraint Order (1/0)	
	(1)
Specialized Services Usage (1/0)	0.176 (0.031)
Local Polynomial	Linear
Time Fixed Effects	Yes
Individual Controls	No
First stage F-statistic p-value	<0.0001
Mean of dependent variable	0.407

Notes: Standard errors clustered at the neighborhood level in parentheses. Local linear fit using Calonico et al. (2016) optimal bandwidth and local linear regression RD estimation.

- For simplicity, assume that the effect of the treatment is homogeneous (constant) across individuals (e.g., $\delta_i = \delta$ for all i .) **Discuss the empirical results.** Your discussion should include [10]:
 - An evaluation of the identification strategy;
 - An explanation and interpretation of how the 2SLS estimates are obtained from the reduced form;
 - A discussion of whether the instrument/s is/are “weak”, and the implications.
 - A substantive discussion of the 2SLS estimates.

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Part C: Relax the homogeneous treatment effects assumption in Part B (above), such that each individual may exhibit a different relationship between specialized services usage and request of and issue of restraint orders against an offender.

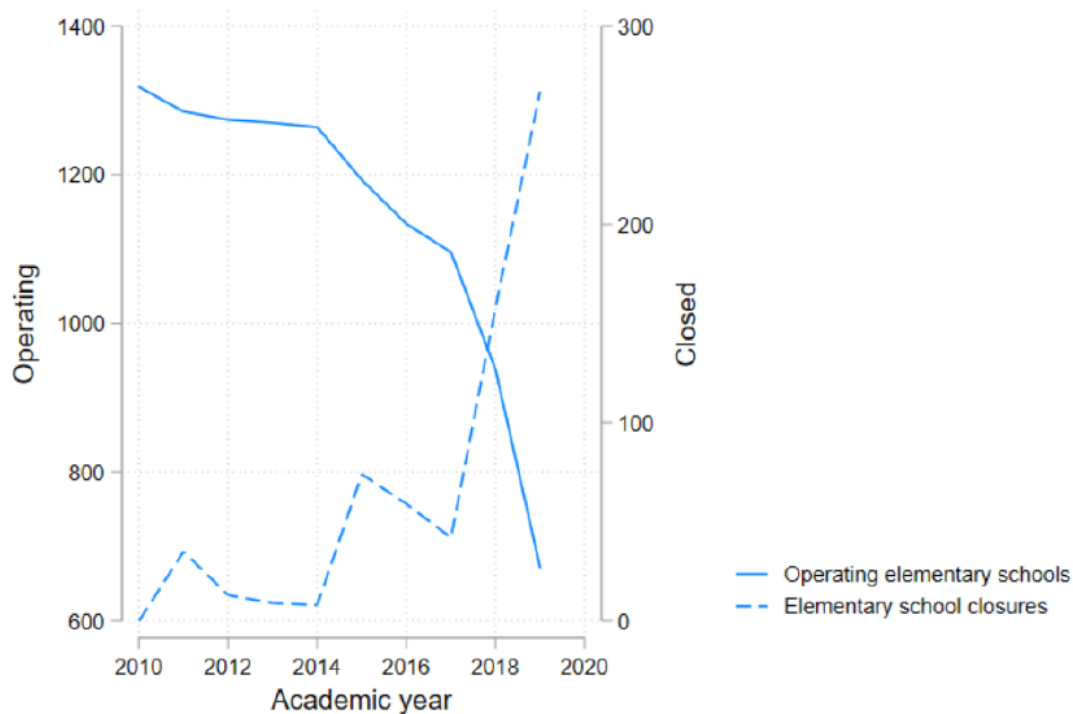
- Evaluate in the context of this application the validity of the key assumptions for the interpretation of the IV estimate as some form of LATE: Independence, Exclusion, First-Stage, and Monotonicity. Interpret the IV estimates in Table 1 under the view of these as some form of LATE [10].

Section 3: Differences in Differences / Event Study and External Effects (20 points)

In the past decades, many school districts across developed countries have been downsizing school availability, due to resource constraints and long-term demographic changes. This has led to policy discussions as to the consequences of such school consolidation policies for students, families, and teachers, as well as debates regarding ways to achieve this that may minimize the disruption possibly caused by such decisions (e.g., Brummet 2014).

A school district in the southern US has implemented such policies during the period 2010-2019, leading to the consolidation of 665 elementary and middle schools (see Figure 3, below).

Figure 3: Operating and Consolidated Elementary and Middle Schools in the School District, by Academic Year



Evaluation: Concerned about the potential adverse consequences of such decisions, the school district's Research and Evaluation Unit (R&E Unit) was tasked with conducting a retrospective evaluation of the consequences of the policy for students' enrollment (drop-out) and academic achievement outcomes.

Data: The R&E Unit has student-level longitudinal administrative data on enrollment and test scores for all children in primary and middle school (during the period 2009-10 to 2017-18) with each school's enrollment classification, open/closed status, and district catchment area to evaluate the reforms.

Part A: The research team would like to use a differences-in-differences / event study design for this evaluation, using the following model structure:

$$y_{igst} = \sum_{k=-4}^{-2} \beta_k D_{i,t}^k + \sum_{k=0}^4 \beta_k D_{i,t}^k + \alpha_i + \pi_t + \delta_{et} \epsilon_{igst}$$

where y_{igst} is the standardized academic achievement outcome of student i in grade g , school s , in academic year t ; α_i is a student fixed effect; π_t are academic year fixed effects. Error terms, ϵ_{igst} , are clustered at the local school district level, to take into account of the serial and cross-sectional correlation of unobserved determinants of achievement outcomes among students. (Ignore the notation δ_{et} – it is a typo in the equation.)

Outcome: The y_{igst} academic achievement outcome is normalized to have mean equal to zero (0) and standard deviation equal to one (1), such that the estimated effects are interpreted as effect sizes.

Treatment: Treatment indicators for each student i are defined as the following indicator variables:

$$D_{i,t}^k = 1\{t - IY_s = k\},$$

the year k before or after initial assignment (IY_s) of the student to a school consolidated at time t . The reference category is year $k = -1$, the year before assignment to treatment.

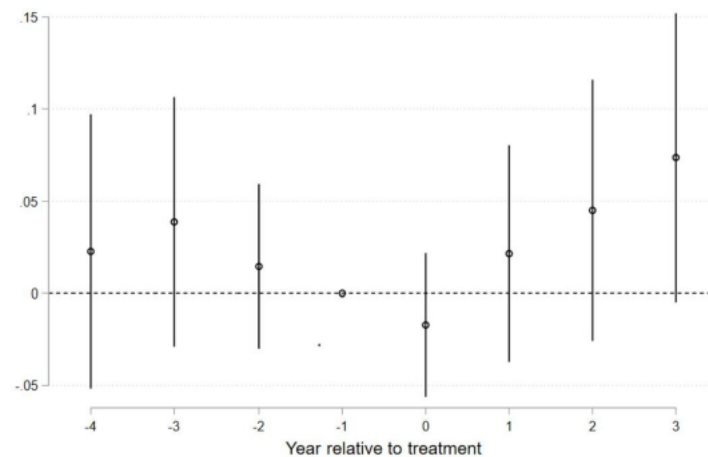
- Given that the model includes student fixed effects, describe the interpretation of the coefficients β_k , $k = \{-4, -3, -2\}$ [5].

Part B

- Given that the model includes student fixed effects, describe the interpretation of the coefficients β_k , $k = \{0, 1, 2, 3, 4\}$ [5].

Part C: The R&E Unit generates the following estimates of the empirical model above reported in Figure 4 (β_k , $k = \{-4, -3, -2, 0, 1, 2, 3\}$) together with 95 percent confidence intervals.

Figure 4: School Consolidation, Displacement of Students, and Average Effects on Math Academic Achievement Trajectories



- Evaluate the difference-in-differences / event study research design and interpret the substantive results of the study [10].

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